INDIA.RUNS

Team Name & ID: team_2+

Offline Candidate Ranking Pipeline

Designed for the 'Senior AI Engineer - Founding Team' Role

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Team Details & Introduction

PRIMARY CONTACT

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- **Role:**
  ML Engineer & Pipeline Developer

DEVELOPMENT FOCUS

- **1. Offline Focus:**
  Engineering highly efficient algorithms designed to process massive candidate datasets locally without network latency.
- **2. Hardware Optimizations:**
  Stream-reading data to prevent out-of-memory errors on typical recruiter computer hardware.
- **3. Hybrid Search:**
  Combining deep semantic transformers with keyword indexes to get explainable shortlists.

SUBMISSION COMPLIANCE

- **1. Team ID Match:**
  Official Team ID is team_2+, matching our registered team name for consistency across portals.
- **2. Metadata Verified:**
  Metadata template fully populated and verified to match official hackathon requirements.
- **3. Output Conformity:**
  All output rankings passed formatting validation. Excel and CSV variants checked.

Problem Statement

THE RECRUITMENT BOTTLENECK

- **1. Volume Overload:**
  Reviewing 100k+ candidate profiles manually is a massive time sink. Recruiters struggle to find qualified profiles in reasonable timelines.
- **2. Keyword Limitations:**
  Traditional applicant tracking systems (ATS) rely on exact keyword matching, filtering out strong candidates who use adjacent terminology.
- **3. Semantic Blindness:**
  Unstructured text summaries in CVs contain massive context that standard relational database filters destroy.

DATA INTEGRITY HAZARDS

- **1. Resume Inflation:**
  Hackathon datasets contain trap profiles (honeypots) that claim impossible timeline metrics or fraudulent skill proficiencies.
- **2. Timeline Discrepancies:**
  Profiles claiming years of experience that exceed their actual years since graduation bypass basic filters.
- **3. Notice Period Agility:**
  Long notice periods lead to severe recruiter drop-offs. Traditional sorting fails to integrate notice constraints dynamically.

Proposed Solution

DECOUPLED TWO-STAGE FUNNEL

- **1. Data Integrity Gate:**
  Programmatic filters stream-read profiles to neutralize fraudulent honeypots before any model scoring.
- **2. Sparse Recall:**
  A fast BM25 lexical index filters 100k down to the top 2,000 candidates in 10 seconds.
- **3. Semantic Re-ranking:**
  Dense vector encoders and Cross-Encoders score semantic fit locally on CPU within minutes.

FUSION OF BEHAVIORAL SIGNALS

- **1. Signal Integration:**
  Integrates notice period constraints, relocation status, and recruiter response rates directly into the ranking equation.
- **2. Notice Agility:**
  Immediate availability triggers ranking bonuses; standard 90-day locks receive minor ranking penalties.
- **3. Activity Weighting:**
  Weights scores based on candidates' active platform engagement and responsiveness metrics.

RECRUITER EXPLAINABILITY

- **1. Factual Reasonings:**
  Generates direct, factual, non-hallucinated explanations for why a candidate is shortlisted.
- **2. Interactive UI Sandbox:**
  Gradio dashboard lets recruiters adjust Job Description in real-time and inspect candidates' score breakdowns.
- **3. Verified Outputs:**
  Enforces strict monotonic ranking format compliance, making the output instantly reliable.

System Architecture

STAGE 1 & 2: DUAL RETRIEVAL

- **1. Streaming Honeypot Filtering:**
  Stream-reads the JSONL candidate dataset record-by-record, keeping memory footprint under 2GB RAM.
- **2. Temporal Integrity Checks:**
  Evaluates total years of experience, skill durations, and job title coherence to filter out 35k trap candidates.
- **3. BM25 Sparse Search Funnel:**
  Uses custom BM25 index over JD keywords to rapidly funnel pool from 64k down to 2,000 in ~10 seconds.

STAGE 3: DENSE EMBEDDING

- **1. Local Dense Encoding:**
  Computes 384-dimensional dense vectors for top 2,000 summaries locally on CPU using SBERT all-MiniLM-L6-v2.
- **2. Semantic Overlap Mapping:**
  Generates cosine similarity matrix against expanded Job Description targets, avoiding simple keyword match constraints.
- **3. Platform Signal Modulation:**
  Fuses Redrob signals (immediate availability bonus, relocation willing, activity) directly into the score math.

STAGE 4: DEEP RE-RANKING

- **1. Cross-Encoder Joint Attention:**
  Loads ms-marco-MiniLM-L-6-v2 locally on CPU to jointly evaluate JD-profile context on the top 200 candidates.
- **2. Joint Relevance Calibration:**
  Generates a sigmoid-scaled relevance rating capturing deep contextual alignments that bi-encoders miss.
- **3. Factual Reasonings Output:**
  Programmatically compiles non-templated rationales detailing years of experience, skills, and notice period constraints.

Core Algorithms & Logic

HONEYPOT IDENTIFICATION ENGINE

- **1. Experience-Graduation Audit:**
  Flagged: Experience > (2026 - first_graduation_year) + 2. This filters profiles containing impossible career timelines.

- **2. Skill Durations Audit:**
  Flagged: 4+ skills claiming 'expert' or 'advanced' proficiency but specifying exactly 0 duration months.

- **3. Career Dates Verification:**
  Flagged: Job duration in months exceeds the dates difference (start_date to end_date) by more than 12 months.

HEURISTIC SCORING MATH

- **1. Notice Period Modifiers:**
  Notice period agility: +5% bonus for <= 15 days notice; -15% penalty for >= 90 days locked periods.

- **2. Experience Fit Curve:**
  Seniority Fit: Underqualified candidates face a steep exponential penalty. Overqualified profiles are decayed slowly.

- **3. Geographic Modulations:**
  Location: 1.0 multiplier for target hubs (Pune, Noida, Delhi NCR). 0.95 for relocation willing. 0.7 for non-willing.

Technologies Used

DEEP LEARNING & NLP WORKloads

- **PyTorch & Transformers:**
  Provides the core tensor computation framework, optimized for local CPU execution.
- **SentenceTransformers (all-MiniLM-L6-v2):**
  Computes fast dense text embedding vectors for candidates summary profiles.
- **CrossEncoder (ms-marco-MiniLM-L-6-v2):**
  Computes joint attention maps across candidates and JD requirements for fine-grained re-ranking.

DASHBOARD & INTERACTION ENGINE

- **Gradio Web App:**
  Powers our interactive, sandboxed web dashboard. Allows real-time JD changes and live shortlist monitoring.
- **Pandas & NumPy:**
  Manages index arrays, matrix similarity multiplications, and candidate data structures.
- **OpenPyXL (XLSX Generator):**
  Compiles the final validated candidate shortlist into spreadsheet format.

LOCAL ARCHITECTURE CHOICES

- **Zero-Network local Cache:**
  Pre-downloaded model weights cached directly in ./models/ for 100% network isolation during ranking runs.
- **Low-Memory Streaming Parser:**
  O(N) streaming JSONL parser ensures memory consumption stays under 2GB, preventing out-of-memory errors.
- **Portable OS Routing:**
  Cross-platform path resolution supports Windows, Linux, and macOS out-of-the-box.

Results & Performance

COMPUTE BENCHMARKS

- **Execution Speed: 193.75 seconds**
  Completes full filtering, BM25 indexing, dense vector similarity, and Cross-Encoder re-ranking in 193.75s.
- **Memory Footprint: <2GB Peak RAM**
  Filters 100k candidates line-by-line using under 2GB peak RAM, ensuring compatibility with standard 16GB CPU laptops.
- **Network Independence: 100% Offline**
  Runs entirely offline with zero API calls, guaranteeing candidate data confidentiality.

RANKING & SCREENING QUALITY

- **Honeypot Screen Rate: 35,208 profiles**
  Screened out 35,208 candidates with fraudulent skills or anomalous timelines, ensuring 0% honeypots in Top 100.
- **Monotonic Order compliance:**
  Ensures scores strictly decrease as rank increases, with deterministic candidate_id sorting breaking duplicates.
- **Validation Suite: 100% Pass**
  Checked and passed by validate_submission.py for strict schema, column ordering, and tie-breaker conformity.

INTERACTIVE WEB APP RUN

- **JD Modulator & UI Shortlist:**
  Allows real-time customization of Job Description and instantly compiles score meters (tech match, behavior).
- **Gradio Sandbox Server:**
  Runs locally on port 7860, with web requests returning HTTP 200.
- **Sandbox Port Verification:**
  Verified by urllib to successfully load and serve profile inspect tabs.

Evaluation & Ranking Quality

SHORTLIST STACK ACCURACY

- **1. Match Quality Alignment:**
  Candidate profiles match the founding AI/ML engineer profile: RAG pipeline experience, vector databases, and evaluation metrics.
- **2. Title Match Calibration:**
  Taxonomy mapping identifies candidates matching 'sde-iii', 'member of technical staff', or 'lead AI engineer' roles correctly.
- **3. Geographic Availability:**
  Re-ranking incorporates candidate relocation readiness to filter out candidates bound to overseas or remote-only modes.

SHORTLIST COHERENCE & TRUST

- **1. Explanations Transparency:**
  Programs generate transparent, non-generic descriptions (e.g. 'Software engineer with 6.9 years experience matching PyTorch, Milvus').
- **2. Factual Compliance:**
  Zero hallucinated text or fabricated skills. Every claim is strictly backed by the candidate's career details.
- **3. Honeypot Screening Proof:**
  Verified to contain 0% honeypots or timeline fraud candidates in the final recommandé Top 100.

Submission Assets

GITHUB REPOSITORY

- **Clean Repository:**
  Contains fully working, clean, and complete code with local model caches and Gradio app.
- **Repository Link:**
  https://github.com/anmolraj499-ai/Candidate-Ranking
- **README Documentation:**
  Detailed instructions on setup, environment, and pipeline execution.

REPRODUCIBILITY SANDBOX

- **Colab Notebook:**
  Allows judges to run candidate evaluation sandbox on Google Colab's standard compute in seconds.
- **Sandbox Link:**
  https://colab.research.google.com/github/anmolraj499-ai/Candidate-Ranking/blob/main/sandbox_ranking.ipynb
- **Automation Workflow:**
  Clones the repo, installs dependencies, and runs ranker on sample_candidates.json.

SHORTLIST OUTPUTS

- **Validated Shortlist CSV:**
  Format-compliant candidate rankings containing candidate_id, rank, score, and reasoning.
- **Excel Shortlist XLSX:**
  Excel XLSX version created for easy upload through the portal's restricted format picker.
- **Submission Metadata YAML:**
  Official metadata detailing compute environment, team contact details, and algorithms used.

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THANK YOU

Team: team_2+